

Plant Station / R6

Concept appearance and illustrated assembly | Seven printed parts plus fit coupon



Updated design: 60 mm-high sump, three 12 mm button bores, 8.4 mm hose clip and an open rear hose/cable route. The planting cavity and measured LCD/ESP32 mounting centres are preserved.

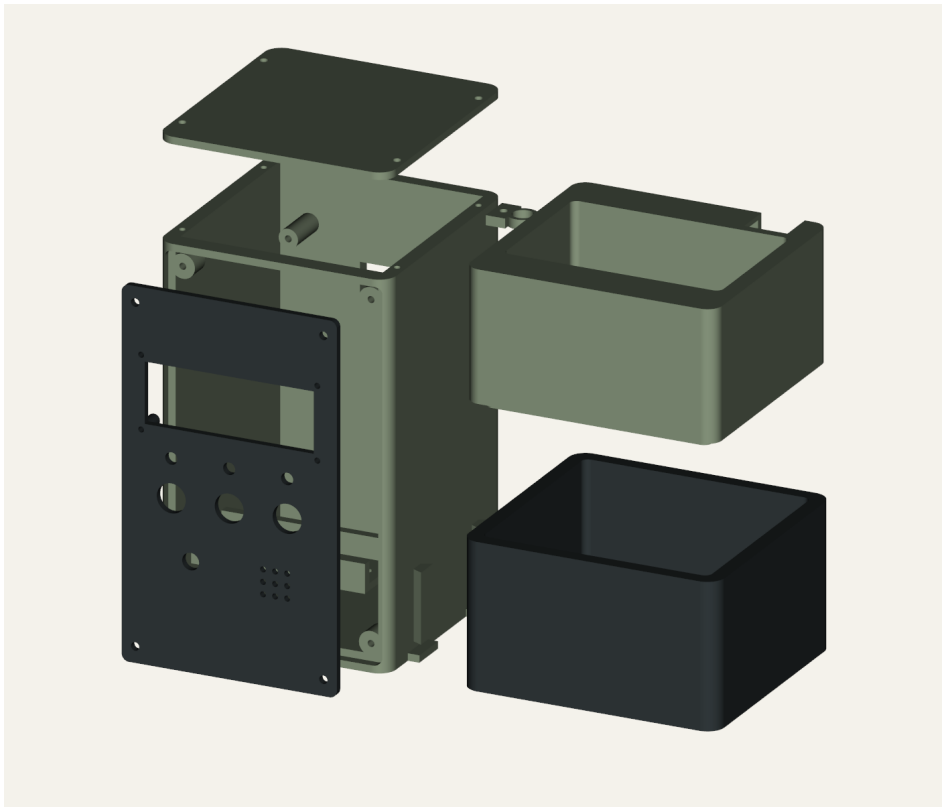
Read the illustration correctly. The plant, LCD text, illuminated button rings, indicator colours and tube shape are illustrative. LCD text output is not implemented in the current firmware. Do not use the picture for dimensions, pin assignments or minimum water level.

Build status: CAD and exported meshes checked. Actual pump, switch nut/body fit, printing process, leak containment and electrical operation remain unverified. Dimensions in the following pages take precedence over appearance proportions.

For an existing R5 build, replace the **sump, planter and hose clip** together. The tower, fascia, top cover and battery tray have unchanged STL hashes.

02 / Parts and arrangement

Assembly geometry from the exported CAD; the exploded spacing is illustrative.



Part / quantity	Role and R5 replacement
tower_sage / 1	Dry enclosure with rear PCB standoffs; retain R5.
fascia_charcoal / 1	LCD, three buttons, LEDs, potentiometer and grille; retain R5.
top_cover_sage / 1	Removable dry-enclosure lid; retain R5.
sump_charcoal / 1	60 mm-high pump reservoir; replace R5.
planter_sage / 1	Removable soil compartment with two drains; replace R5.
battery_tray / 1	Insulated-holder support; retain R5.
hose_clip / 1	8.4 mm bore, external routing; replace R5.
button_fit_coupon / 1 test piece	12.0 / 12.2 / 12.4 mm holes; not part of final assembly.

Import assembly STLs as **millimetres**, keeping common coordinates and disabling auto-arrange for inspection. For printing, orient each part and place it on the bed. No slicer profile or universal support/temperature settings are specified.

03 / Interfaces and fit coupon

Datum: x right, y rear, z up; origin at the tower front-left-bottom corner.

Interface	Nominal geometry	Evidence / allowance
Overall assembly	207 x 100 x 161	CAD envelope, excluding plant and external tubing
Sump main body	106 x 100 x 60; floor 4	56 internal depth; footprint excludes dovetail collar
Planting cavity	90 x 60 x 50; two drains dia. 6	CAD nominal; not a recommended fill volume
Button bores	3 x dia. 12.0, 24 mm centres	User-specified holes; fascia thickness 3
Button rear space	Nut dia. 20; rear depth 35	Assumed: first 5 mm nut space, then dia. 16
LCD	70 x 25 aperture; 73 x 30 centres	Preserved user measurements
ESP32 carrier	60 x 55 mounting centres	Preserved user measurements
Hose clip	8.4 bore for nominal 8 OD tube	0.4 diametral allowance; retention untested

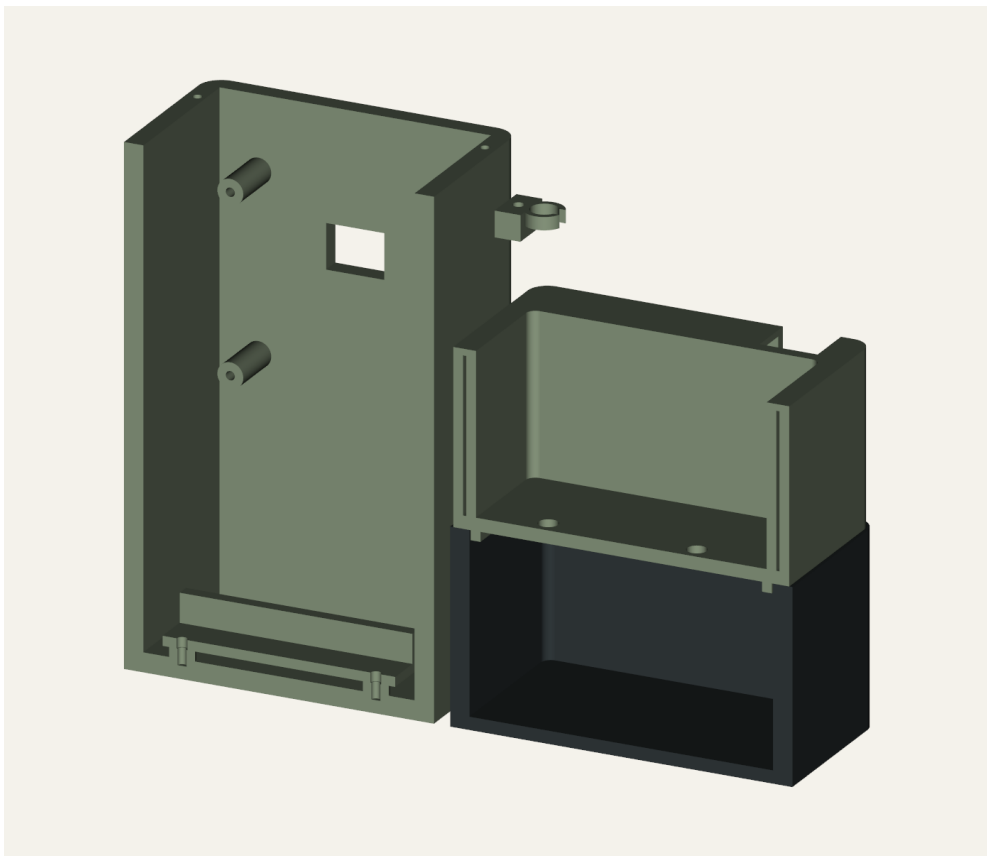
Step 1 - Establish insert fit. Print the 3 mm-thick coupon first. The corner notch marks the 12.0 mm end; the next holes are 12.2 and 12.4 mm. Test the actual threaded body, sealing washer and nut. The fascia remains 12.0 mm nominal: do not scale the complete fascia to compensate for one hole.

Step 2 - Check rear space. Measure body depth including terminals and wire bends, nut diameter, permitted panel thickness and thread engagement. Confirm the nut tool can reach without disturbing adjacent inserts. If the 20 mm/35 mm envelopes are exceeded, revise the CAD before full printing.

Photo limitation: the purchase screenshot shows red/green/yellow ring styles but not the selected momentary/latching action or LED voltage. Those are electrical identification checks, not facts established by the 12 mm hole size.

04 / Assemble the wet module

Keep electrical parts disconnected during dry fitting and water-containment checks.



Step 3 - Trial-fit the pump. Reference body: 38.5 L x 25.5 W x 43 H mm, upright on the sump floor. CAD checks include 3 mm lateral/top allowance. The body has 13 mm to the main planter underside; the peripheral lip level is 10 mm above it. Add the real outlet, feet and cable bends to the envelope.

Step 4 - Route the tube. Fit tubing to the pump outlet and lead it through the 18 mm rear service notch. Route the cable beside it without pinching. The open notch is outside the soil cavity; protect its edges and maintain a drip loop before any dry-enclosure entry. Fit the external clip without crushing the tube.

Step 5 - Make it serviceable. Use actual pump-compatible removable retention and inlet screening. No specific suction-foot mount is modelled. Fit removable debris guards over both drains; keep both paths open and accessible. A liner or coating must not close them.

Step 6 - Close and test. Lower the planter onto its locating lip with no trapped hose/cable. Test the empty wet module over a tray for leaks and drain-back overflow before adding soil or electronics. Establish a measured maximum fill level below the soil floor and an operating minimum above the pump requirement.

Step 7 - Couple the modules. Slide the sump collars onto the tower dovetails from above, keeping the bases level. Do not force the joint. It is not a carrying lock: support both modules when moving the assembly.

05 / Assemble the dry module

Fit components with the fascia and top cover removed; all power disconnected.

Step 8 - Fit the LCD to the fascia. Use the four 2.4 mm bores and 8 mm standoffs. M2 through-fixings are provisional; keep screw-head diameter at or below 4 mm near the aperture. Select length from the actual stack and nut engagement. Do not enlarge bosses or bow the LCD PCB.

Step 9 - Fit the controls. Insert the three 12 mm buttons from the front; install their supplied washers/nuts inside. Add LEDs, potentiometer and sounder using measured fittings. Do not apply a guessed metal-panel tightening torque to printed plastic. Keep the wiring serviceable.

Step 10 - Fit the board and holder. Check the SunFounder revision and underside components before using the 60 x 55 mm mount. Fit an insulated cell holder to the tray, then the tray to the floor pads; the tray is not a bare-cell contact system. Leave the cell disconnected.

Step 11 - Arrange the harness. Keep pump current wiring out of the GPIO circuit and separated from the wet module. Provide strain relief and insulating mounts for any driver/level-shifter boards. Check real PCB/LCD projections; the existing envelope assumptions do not include every connector.

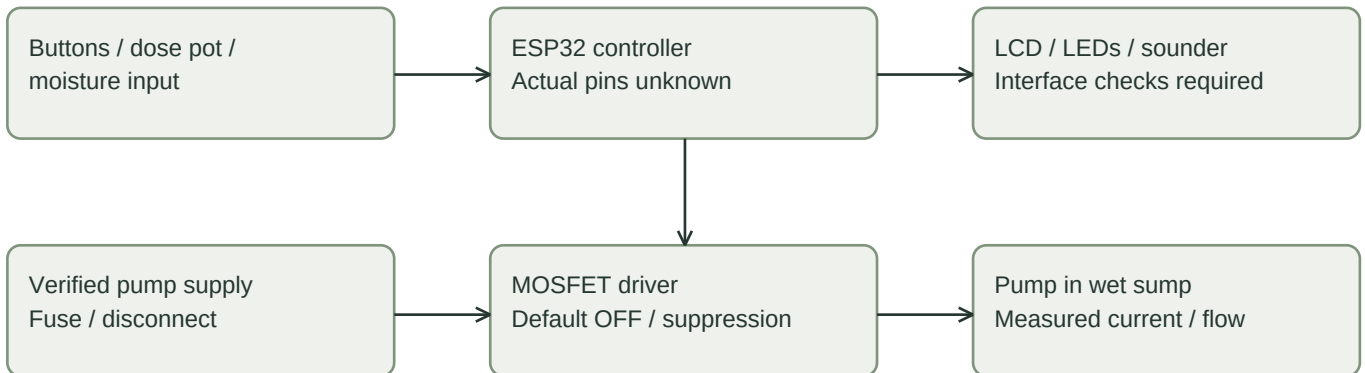
Step 12 - Close for inspection. Fit the fascia and top cover only after checking wiring clearance, USB cable insertion, switch nut clearance and fastener length. Open the covers again if a wire is compressed. Confirm both covers can be removed for service.

Fastening point	CAD geometry / practical check
LCD x4	2.4 through-bore; provisional M2; maximum head dia. 4
Rear board x4	2.8 pilot; confirm screw and board bore; do not bottom out
Fascia x4 / lid x4	3.4 clearance / 2.8 pilot; select length from actual engagement
Holder tray x2 / hose clip x1	3.4 clearance / 2.8 pilot; clip blind pilot is only 3 deep

No heat-set inserts or validated installation torques are specified. Start screws square and verify retention on a separate printed sample. The existing LCD boss-to-aperture margin is small (approximately 0.415 mm).

06 / Electrical assembly boundary

Logical relationships only - this is not a physical SunFounder pinout or a ready-to-wire schematic.



Step 13 - Identify before wiring. Match the carrier revision, reserved camera/SD pins, cell and connectors. The candidate carrier includes charging and a 5 V converter, but allowable pump startup load and simultaneous USB/battery operation are unresolved. Do not use the classic DevKit compile-review pin map as a physical wiring plan.

Step 14 - Verify interfaces. Keep ESP32 signals within their 3.3 V limits. If the LCD backpack pulls I2C to 5 V, fit suitable bidirectional translation. Each separate LED needs its own resistor. Verify the purchased button-ring LED voltage/polarity and switching action. Supply a sounder through an appropriate driver.

Step 15 - Verify pump shutdown. Use the measured pump voltage and startup current to select supply, driver, suppression, fuse and cable. A 4.5 V nominal pump listing is not proof of 5 V compatibility. Check default-OFF hardware and common-ground routing against the final schematic; never drive the pump from GPIO.

Current firmware: manual timed dose only. WATER captures the dose setting once; releasing WATER does not end that dose. STOP and LAMP TEST cancel it. Release/re-arm is required. Moisture is advisory. Duration is not calibrated volume. LCD text, button illumination and actual hardware operation are not demonstrated by the concept art.

Start with unpowered assembly checks. Keep the pump disconnected until board-specific configuration and dry output tests are complete. No upload or powered test is performed by this documentation update.

07 / Acceptance worksheet

Record evidence against the actual built unit. An unchecked row remains open.

Unit / build ID: _____ Date: _____
Pump model: _____ Switch variant: _____
Printer / material / profile: _____

Check	Acceptance / evidence to record	Result
A1 Insert coupon	Body fits; nut/washer seats on 3 mm panel; thread and tool access adequate	_____
A2 Pump envelope	Body, outlet, feet and cable clear planter and notch without force	_____
A3 Reservoir	Leak test method/duration recorded; no leakage; drains clear	_____
A4 Water levels	Measured max fill permits drain-back; operating min satisfies pump	_____
A5 Tube route	No kink/pinch; barb and clip retain tube; outlet stays above soil	_____
A6 Dry enclosure	No crushed wires; insulated mounts; USB and cover access retained	_____
A7 Board/switch ID	Carrier pins, button action and ring LED ratings verified	_____
A8 Dry output test	Reset/boot/STOP/lamp test do not create unintended pump command	_____
A9 Supervised wet test	Startup voltage/current and driver temperature acceptable to ratings	_____
A10 Dose calibration	Installed flow and repeatability measured; no assumed ml scale	_____

Measured values

Pump running / startup current: _____ / _____ A
Supply minimum during startup: _____ V
Maximum / minimum reservoir level datum: _____
Dose duration / delivered volume / repeats: _____

Outstanding defects / corrective action: _____

Checked by: _____ Date: _____

08 / Evidence and revision record

Document R6-A1; assembly instructions supplement the retained source files.

Evidence	Recorded result / limitation
R6 CAD	Seven valid solids; 21 assembly pair checks with zero overlap to 0.00001 mm3; positive overlap control 500 mm3
Exported meshes	Eight STLs: closed, consistent winding, one component, positive volume, two faces per edge, no degenerate faces
Interface checks	Button/coupon exported cross-sections; sump height; CAD pump/button envelopes, passage and mount probes
Visual checks	CAD assembly/section and generated appearance reviewed; appearance is not metrology
Runtime limitation	CAD builder printed PASS then returned code 1 on shutdown; independent mesh checker returned success
Not established	Physical fit, self-intersection exhaustion, slicer/material/process approval, leak strength, electrical operation

Controlled references

- R6 dimension contract and assembly notes
- CAD verification record
- Independent mesh verification record
- Editable CAD source
- Firmware scope and commissioning limits
- Electronics design notes

Revision R6-A1 / 11 September 2026. Added generated R6 concept art, current part list, illustrated wet/dry assembly sequence, insert coupon guidance, logical electrical boundary and acceptance worksheet. No STL geometry or firmware changes. This guide supersedes R5 mechanical assembly illustrations; retain R5 for history.

Source hierarchy: user measurements and exact component drawings, then controlled CAD and verification records, then assembly guidance. Appearance art is last. Record any measured deviation before altering the model. The art was generated with the built-in image tool from the R6 CAD assembly reference; prompt and review notes are stored with the art.